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ANTIBIOTIC DERIVATIVES OF KANAMYCIN

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No Drawing. Continuation-in-part of abandoned application Ser. No. 162,315, July 13, 1971. This application Jan. 27, 1972, Ser. No. 221,378

Int. Cl. C07c *47/18*

U.S. Cl. 260—210 AB 13 Claims

ABSTRACT OF THE DISCLOSURE

Derivatives of kanamycin A and B have been prepared which possess substantially improved antibacterial activity. An example of such an agent is 1-[L-(—)-γ-amino-α-hydroxybutyryl]-kanamycin A [IVa, BB-K8].

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation-in-part of a copending application, Ser. No. 162,315, filed July 13, 1971, and now abandoned.

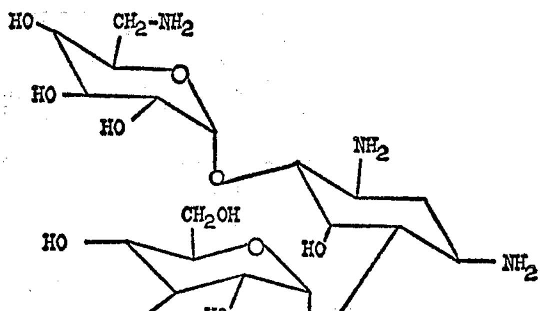
BACKGROUND OF THE INVENTION

(1) Field of the invention

This invention relates to a semisynthetic 1-substituted derivatives of kanamycin A or B, said compounds being prepared by acylating the 1-amino-function of kanamycin A or B with a γ-amino-α-hydroxybutyryl moiety.

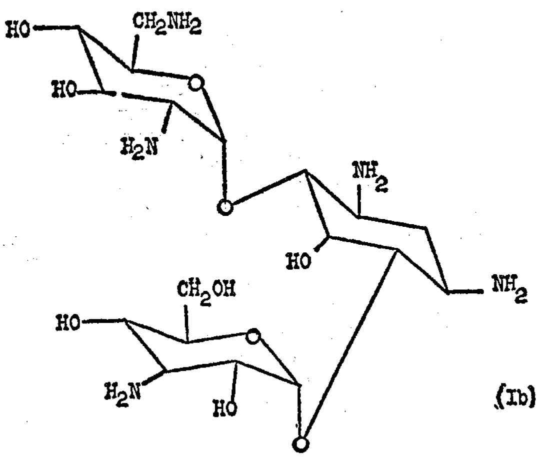
(2) Description of the prior art

The kanamycins are known antibiotics described in Merck Index, 8th ed., pp. 597–598. Kanamycin A is a compound having the formula



(Ia)

Kanamycin B is a compound having the formula



(Ib)

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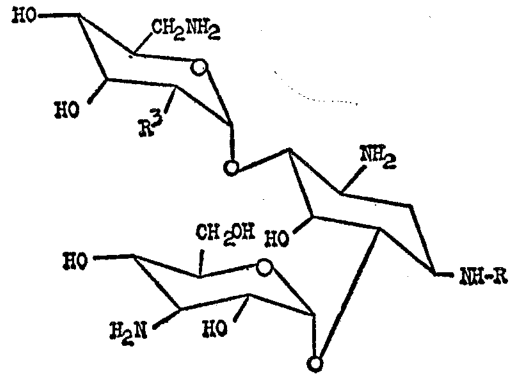
SUMMARY OF THE INVENTION

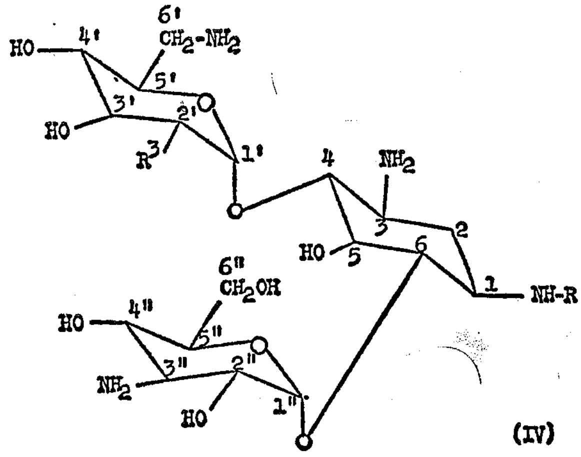
The compound having the formula



(IV)

in which R^3 is OH or NH_2 and R is L-(—)-γ-amino-α-hydroxybutyryl; or a nontoxic pharmaceutically acceptable acid addition salt thereof is a valuable antibacterial agent.

This invention relates to semi-synthetic derivatives of kanamycin A and B, said compounds being known as 1-[L-(—)-γ-amino-α-hydroxybutyryl]-kanamycin A and B and having the formula



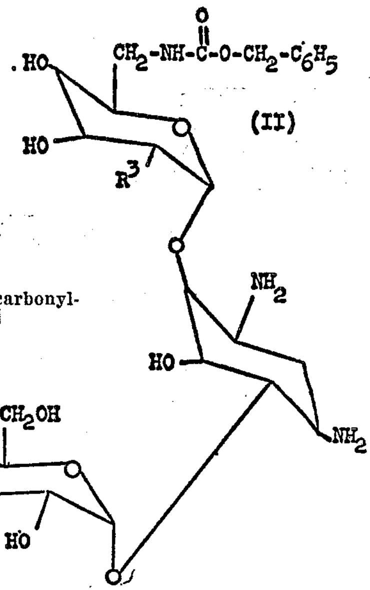
(IV)

in which R^3 is OH or NH_2 and R is L-(—)-γ-amino-α-hydroxybutyryl; or a nontoxic pharmaceutically acceptable acid addition salt thereof.

For the purpose of this disclosure, the term "nontoxic pharmaceutically acceptable acid addition salt" shall mean a mono-, di-, tri- or tetrasalt formed by the interaction of 1 molecule of compound IV with 1–4 moles of a nontoxic, pharmaceutically acceptable acid. Included among these acids are acetic, hydrochloric, sulfuric, maleic, phosphoric, nitric, hydrobromic, ascorbic, malic and citric acid, and those other acids commonly used to make salts of amine containing pharmaceuticals.

The compounds of the present invention are prepared by the following diagrammatic scheme:

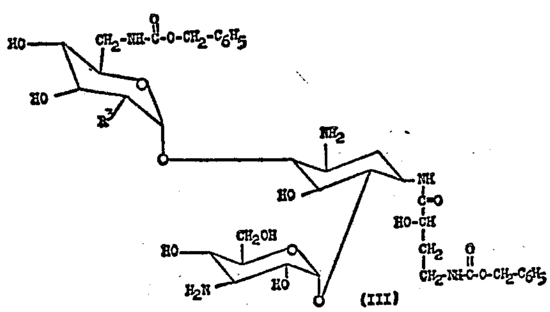
(1) Kanamycin A (Ia) or Kanamycin B (Ib) $\xrightarrow[\text{Succinimide}]{\text{N-(benzyloxycarbonyloxy)}}$



(II)

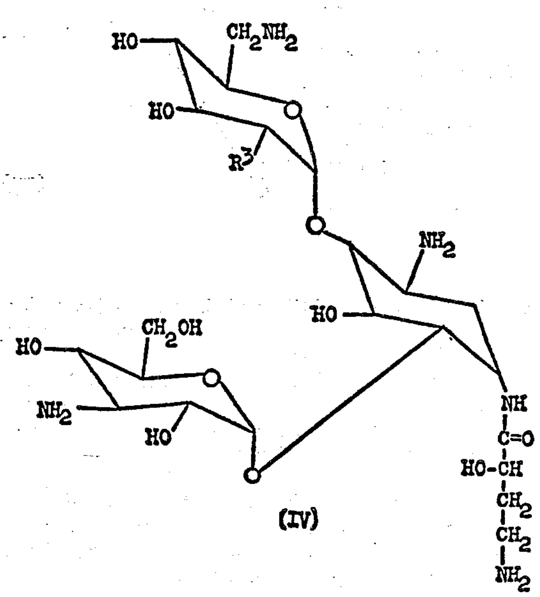
[6'-monobenzyloxycarbonyl-kanamycin A or B]

(2) Compound II $\xrightarrow{\text{N-hydroxysuccinimide ester of L-(—)-γ-benzyloxycarbonylamino-α-hydroxybutyric acid}}$



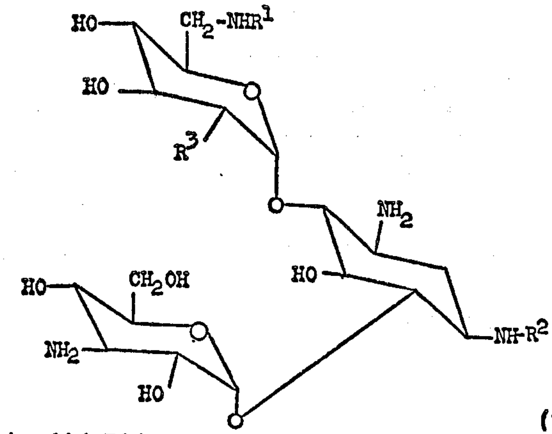
(III)

(3) Compound III $\xrightarrow{H_2/Pd/C}$



(IV)

A preferred embodiment of the present invention is the compound having the formula



(V)

in which R^1 is

$$H \text{ or } C_6H_5-CH_2-O-\overset{O}{\overset{\|}{C}}-$$

R^2 is H or L-(—)-γ-amino-α-hydroxybutyryl and R^3 is OH or NH_2 wherein R^1 or R^2 must be other than H; or a nontoxic pharmaceutically acceptable acid addition salt thereof.

Another preferred embodiment is the compound of Formula V in which R^1 is

$$C_6H_5CH_2O-\overset{O}{\overset{\|}{C}}-$$

R^2 is H and R^3 is OH or NH_2.

A more preferred embodiment is the compound of Formula V in which R^2 is L-(—)-γ-amino-α-hydroxybutyryl, R^1 is hydrogen and R^3 is OH or NH_2.

A most preferred embodiment is the compound of Formula V wherein R^1 is H, R^2 is L-(—)-γ-amino-α-hydroxybutyl and R^3 is OH; nontoxic pharmaceutically acceptable acid addition salt thereof.

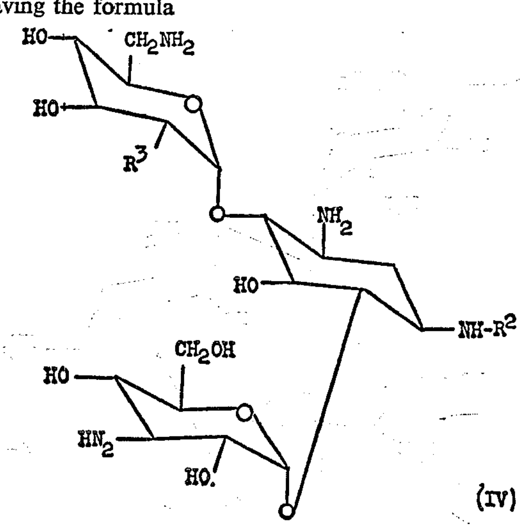
Another most preferred embodiment is the compound of Formula V wherein R^1 is H, H^2 is L-(—)-γ-amino-α-hydroxybutyryl and R^3 is NH_2; or a nontoxic pharmaceutically acceptable and addition salt thereof.

Other most preferred embodiments are the sulfate, hydrochloride, acetate, maleate, citrate, ascorbate, nitrate or phosphate salts of compound V.

Another most preferred embodiment is the monosulfate salt of compound V.

Still another preferred embodiment is the disulfate salt of compound V.

The objectives of the present invention have been achieved, by the provision according to the present invention of the process for the preparation of the compound having the formula



(IV)